

Phase 8: Project Demonstration — Scalability Roadmap & Future Expansion Plan

Project Name: Cab Booking (UCab)

Project ID: N/A (Solo Track Submission)

Team Member / Solo Developer: Shaik Sumiya Zainab

1. Future Scalability Architecture Plan

While Phase 1 delivers a feature-complete and robust MVP capable of supporting hundreds of concurrent simulated trips, expanding UCab into a multi-city commercial enterprise requires specific architectural upgrades across Phase 2 and Phase 3.

```

timeline
title UCab Scalability & Evolution Roadmap
Phase 1 : Core MVP (Current Delivery)
: MERN Monolith on Port 8000
: Zero-Config Dual Database Storage
: Simulated GPS Progress Bar
: QR Corporate Thermal Invoices
Phase 2 : Real-Time & Payment Webhooks (Months 1-3)
: Bi-Directional WebSocket Driver Streaming (Socket.io)
: Live Stripe & Razorpay Webhook Checkout
: Driver Partner App & Live GPS Push
Phase 3 : Distributed Microservices & AI (Months 4-6)
: Microservice Split (Auth, Fleet, Dispatch, Billing)
: Redis Caching for Sub-Millisecond Fleet Lookup
: AI/ML Surge Pricing & Demand Prediction via GeoJSON

```

2. Technical Roadmap Breakdown

Phase 2: Real-Time Telemetry & Production Payment Gateway (Next 90 Days)

- 1. WebSocket / Socket.io Integration: Replace the stateful polling/simulation tracking bar in UserHome.jsx with bi-directional WebSocket channels (ws://localhost:8000). Drivers will push live GPS coordinates every 3 seconds, rendering an animated marker on a Mapbox/Google Maps canvas.
- 2. Stripe & Razorpay Webhook Verification: Connect bookingController.js directly to Stripe payment intents, verifying card verification cryptograms and webhook events (payment_intent.succeeded) before marking trips as Accepted.

Phase 3: Distributed Microservices & AI/ML Forecasting (Months 4 to 6)

- 1. Microservice Decomposition: Split the single Express API server (server.js) into four isolated Dockerized microservices:
 - ucab-auth-service: Handles JWT issuing, OAuth, and Bcrypt validation.
 - ucab-fleet-service: Manages vehicle inventory, image attachments, and categorization.
 - ucab-dispatch-service: Handles trip matching and driver assignment.
 - ucab-billing-service: Generates QR invoices and manages tax compliance.
- 2. Redis In-Memory Caching: Introduce Redis (Redis Cloud / ElastiCache) in front of /api/cars to cache available fleet inventory, reducing database query latency from 12ms down to < 1ms during peak city rush hours.
- 3. AI/ML Demand Surge Forecasting: Integrate a Python machine learning service (trained on historical trip coordinates) to predict surge demand across specific urban geofences, adjusting per-KM rates dynamically to incentivize drivers to move into high-demand zones.